

MATHEMATICS 5310

Hints and solutions to selected exercises from 2.4

2. It does not satisfy the symmetric relation (resp. the transitive relation). EG it is not true that $1 \sim 1$ (resp. $1 \sim 2$ and $2 \sim 1$ but it is not true that $1 \sim 1$).

3. Given $a \in S$, there may be no element b such that $a \sim b$.

4. Define $a \sim b$ if and only if there exists an α such that $a \in S_\alpha$ and $b \in S_\alpha$

5. See Quiz 5.

6. $H = \{i, (1, 2)\}$. Right cosets: $Hi = H(1, 2) = \{i, (1, 2)\}$, $H(1, 3) = H(1, 3, 2) = \{(1, 3), (1, 3, 2)\}$ and $H(2, 3) = H(1, 2, 3) = \{(2, 3), (1, 2, 3)\}$. Left cosets: $iH = (1, 2)H = \{i, (1, 2)\}$, $(1, 3)H = (1, 2, 3)H = \{(1, 3), (1, 2, 3)\}$ and $(2, 3)H = \{(2, 3), (1, 3, 2)\}$.

7. No.

8. Let $a \in G$, then $aH = Hb$ for some $b \in G$. Since $a \in aH$, we have $a \in Hb$ and hence $Hb = Ha$. But then $aH = Ha$ and so $aHa^{-1} = H$ (easy check).

9. $[0] + H, [1] + H, [2] + H$ and $[3] + H$.

10. $i_{\mathbb{Z}_{16}}(H) = 4$.

11. The number of left and right cosets is computed by $m(G)/m(H)$.

12. No: in S_3 we have $H(1, 3) = H(1, 3, 2)$ but $(1, 3)H \neq (1, 3, 2)H$.

13. $U_{18} = \{1, 5, 7, 11, 13, 17\}$ $o(1) = 1, o(3) = 6, o(7) = 3, o(11) = 6, o(13) = 3, o(17) = 2$. U_{18} is cyclic.

14. $U_{20} = \{1, 3, 7, 9, 11, 13, 17, 19\}$. $o(1) = 1, o(3) = 4, o(7) = 4, o(9) = 2, o(11) = 2, o(13) = 4, o(17) = 4, o(19) = 2$. U_{20} is not cyclic.

15. If $x \neq \pm 1$, then $x + 1$ and $x - 1$ are invertible in $\mathbb{Z}_p$. If $x^2 = 1$, then $(x - 1)(x + 1) = x^2 - 1 = 0$ so that $1 = (x + 1)^{-1}(x - 1)^{-1}0 = 0$ which is impossible.

16 and 17. If $a_i \neq a_i^{-1}$, then $a_i = a_j$ for some $i \neq j$ hence we may assume that x is a product of those a_i such that $a_i^2 = e$. Clearly then $x^2 = e$. If $a_i^2 = e$, then the order of the cyclic group $\langle a_i \rangle$ is 2 and it divides the order of G . Therefore, if the order of G is odd, $x = e$.

17. Since $(p - 1)!$ is the product of the elements of the group $\mathbb{Z}_p^*$, by Ex 15, $(p - 1)! = \pm 1$ in $\mathbb{Z}_p^*$ i.e. $(p - 1)! \equiv \pm 1 \pmod{p}$. But by the proof of 16, $(p - 1)!$ is the product of the elements of order 2 in $\mathbb{Z}_p^*$ i.e. $(p - 1)! = 1 \cdot (-1) = -1$ in $\mathbb{Z}_p^*$.

19. $\{e\}, \{(1, 2), (1, 3), (2, 3)\}, \{(1, 2, 3), (1, 3, 2)\}$.

20. $(T_{c,d})^{-1} = T_{c^{-1}, -dc^{-1}}$ and so $T_{c,d}^{-1}T_{a,b}T_{c,d} = T_{c^{-1}, -dc^{-1}}T_{ac, ad+b} = T_{c^{-1}ac, c^{-1}(ad+b)-dc^{-1}} = T_{a, c^{-1}(d(a-1)+b)}$. One can check that the conjugacy class of $T_{a,b}$ is $\{T_{a,\beta} | \beta \in \mathbf{R}\}$.

21. $G = \{e, g, g^2, g^3, f, fg, fg^2, fg^3\}$ and $f^2 = g^4 = 1$ while $fg = g^3f$. The conjugacy classes are $\{e\}$, $\{g^2\}$, $\{g, g^3\}$, $\{f, fg^2\}$, $\{fg, fg^3\}$.

23. $9^2 = 81 = -1$.

24. $p - 1 = 2(2n + 1)$. If $x^2 \cong -1$, then $x^{p-1} \cong (-1)^{2n+1} \cong -1$ contradicting Fermat's Theorem.

25. If n is not prime, then $n = a \cdot b$ for some $0 < a, b < p$, but then $[a][b] = [0]$ and so $[a]$ and $[b]$ have no inverses. Thus $\mathbf{Z}_n^*$ is not a group.

If n is prime, then $(n, a) = 1$ for any $0 < a < p$ and so we may find $p, q \in \mathbf{Z}_n$ such that $pn + qa = 1$. But then $[a] \cdot [q] = [aq + pn] = [1]$. Therefore every element in $\mathbf{Z}_n$ has an inverse. The other axioms are easy to verify.